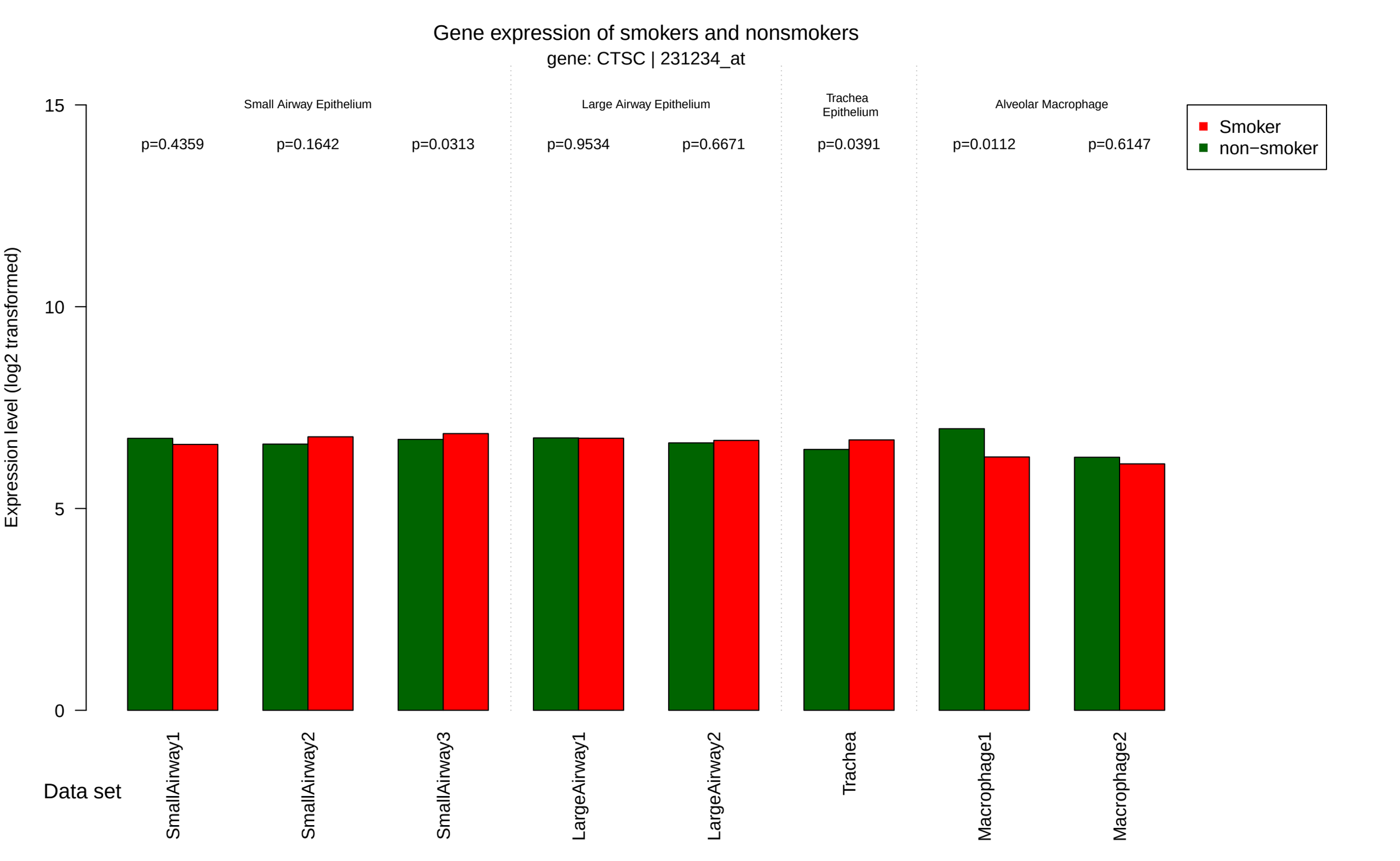
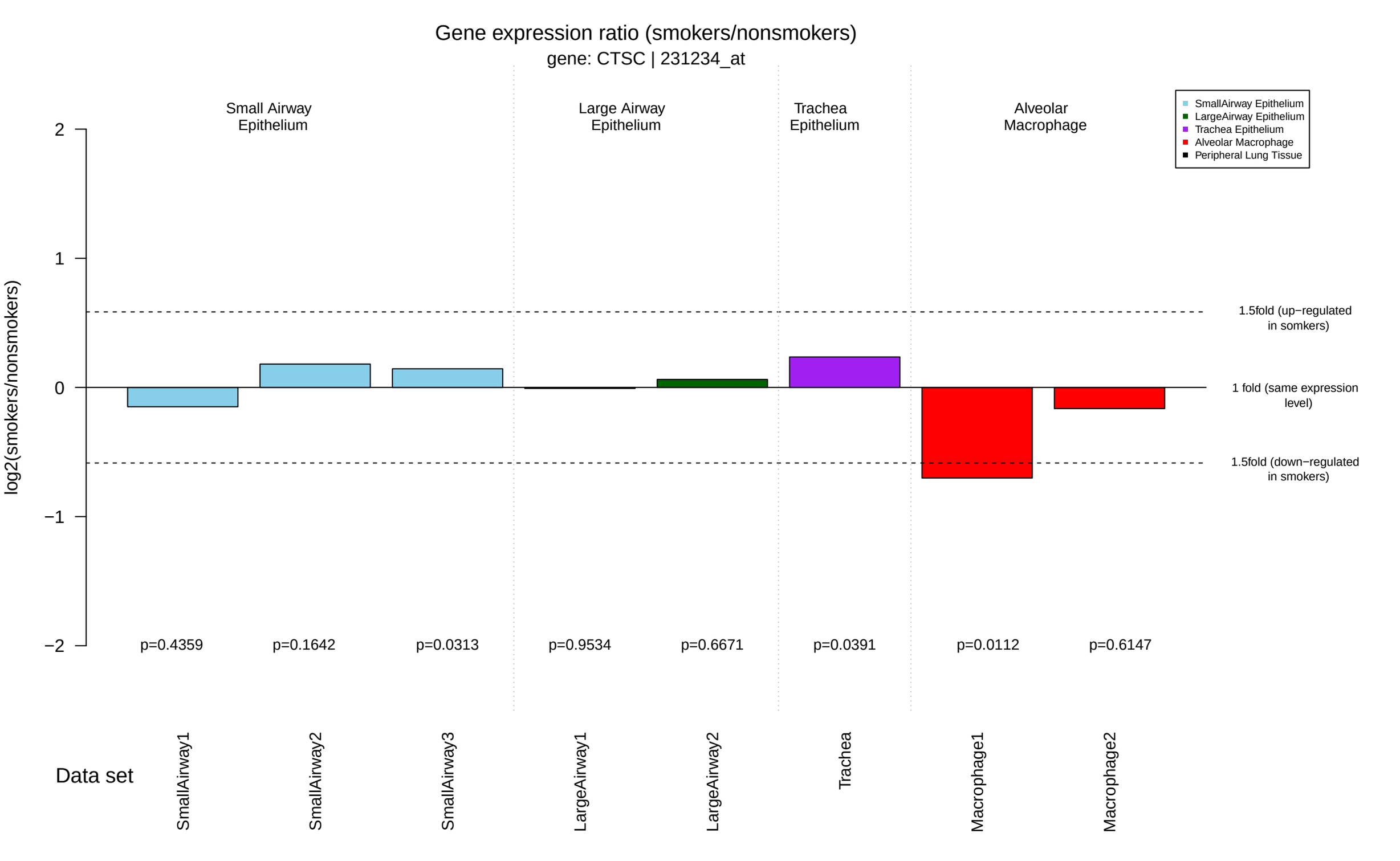
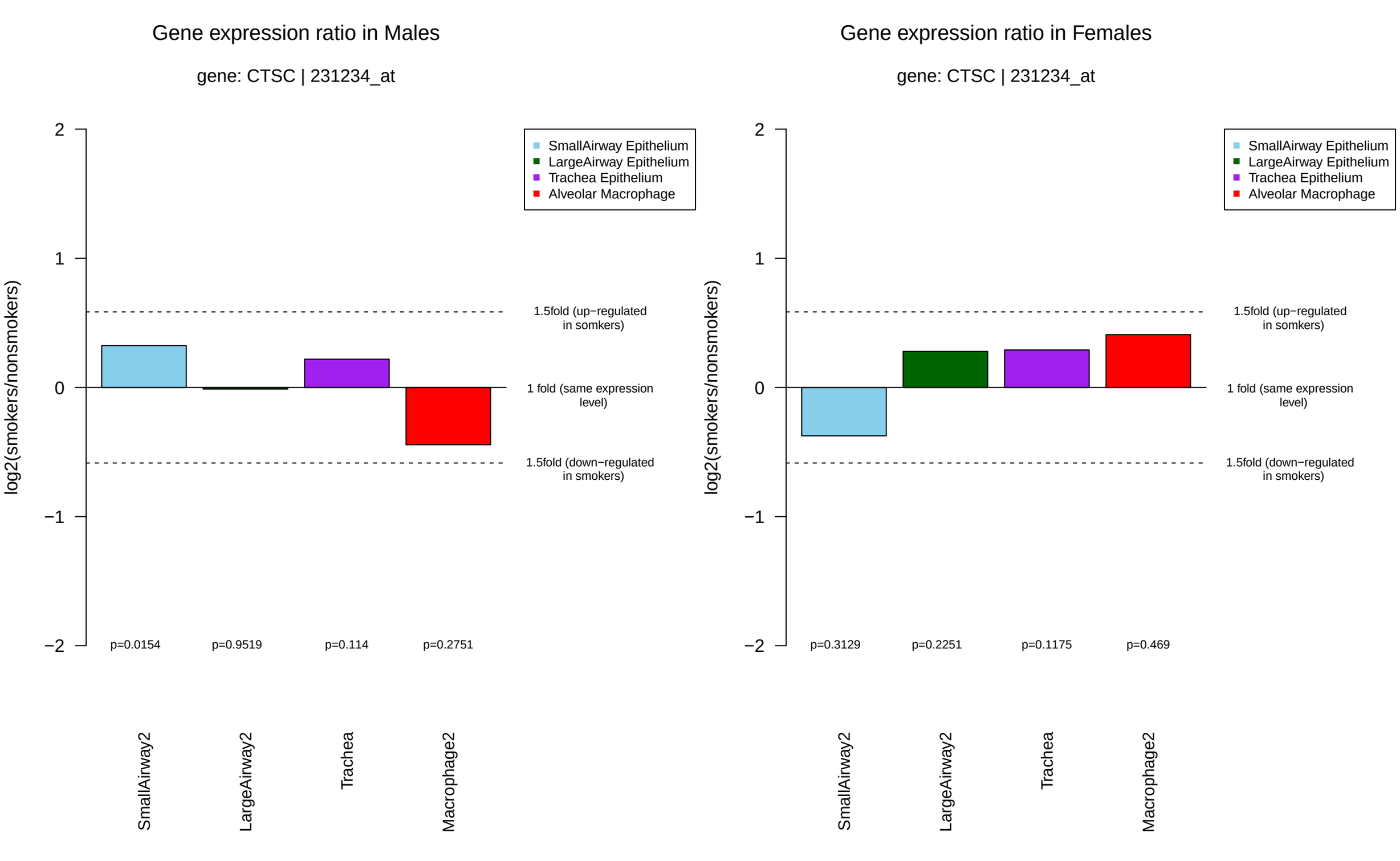
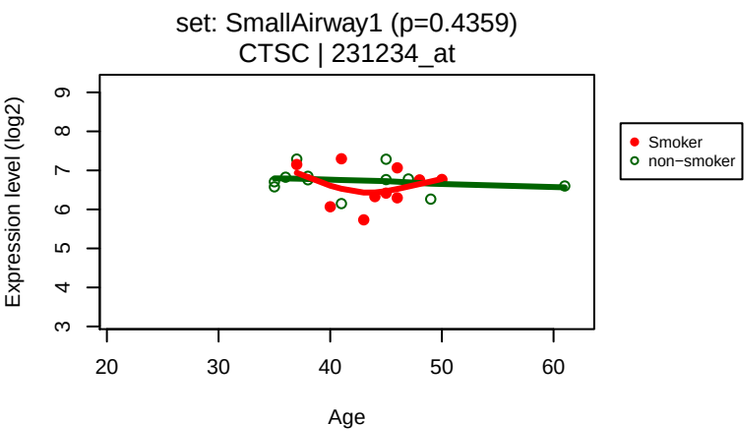
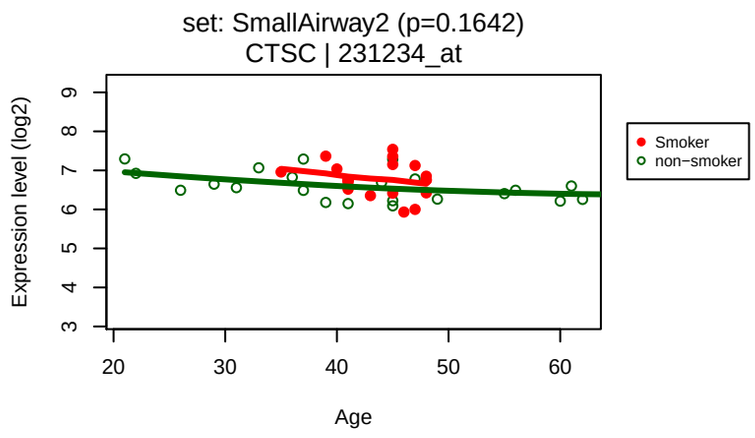
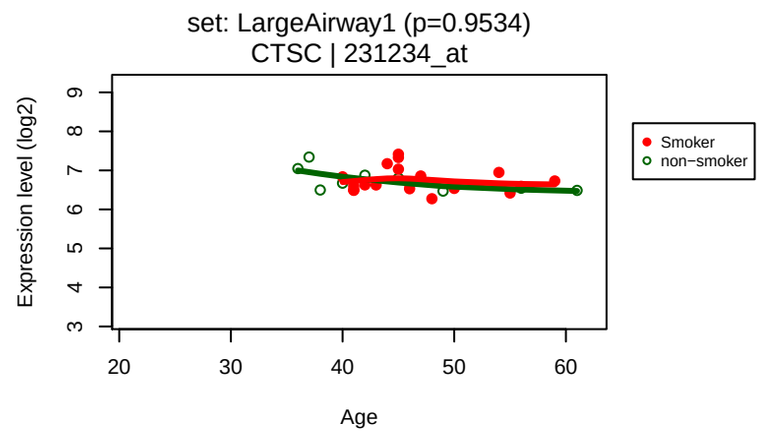

Gene expression of smokers and nonsmokers
gene: CTSC | 231234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4359
p=0.1642
p=0.0313
p=0.9534
p=0.6671
p=0.0391
p=0.0112
p=0.6147
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

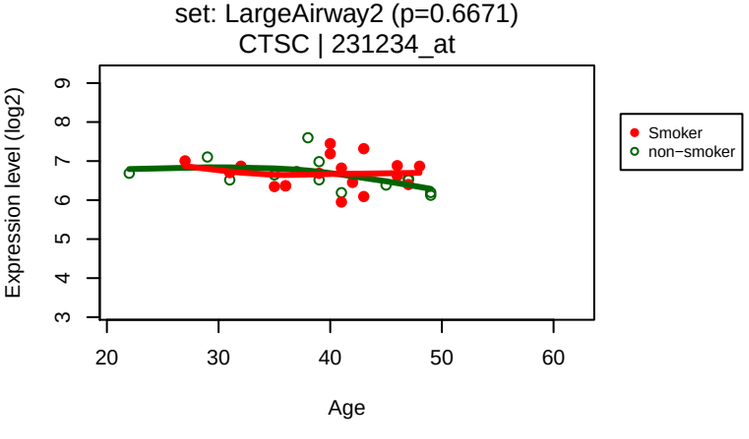

Gene expression ratio (smokers/nonsmokers)
gene: CTSC | 231234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4359
p=0.1642
p=0.0313
p=0.9534
p=0.6671
p=0.0391
p=0.0112
p=0.6147
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

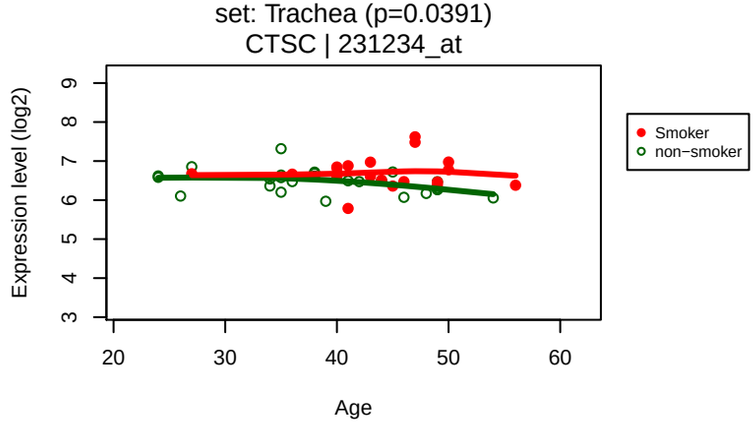

Gene expression ratio in Males
gene: CTSC | 231234_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0154
p=0.9519
p=0.114
p=0.2751
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CTSC | 231234_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3129
p=0.2251
p=0.1175
p=0.469
SmallAirway2
LargeAirway2
Trachea
Macrophage2

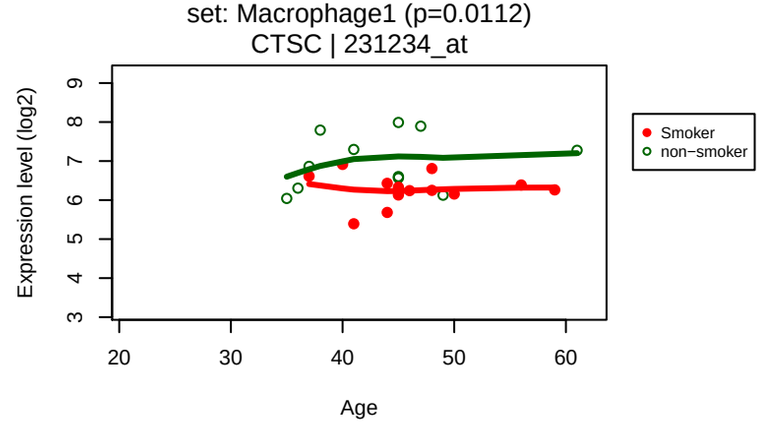

set: SmallAirway1 (p=0.4359)
CTSC | 231234_at
Expression level (log2)
Age
Smoker
non−smoker

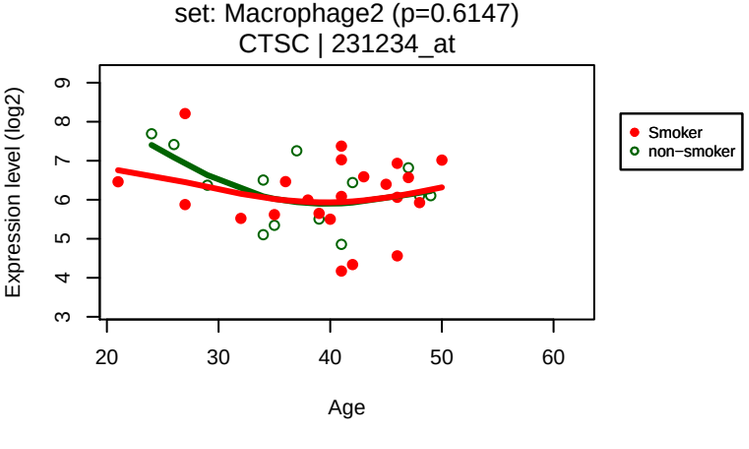

set: SmallAirway2 (p=0.1642)
CTSC | 231234_at
Expression level (log2)
Age
Smoker
non−smoker

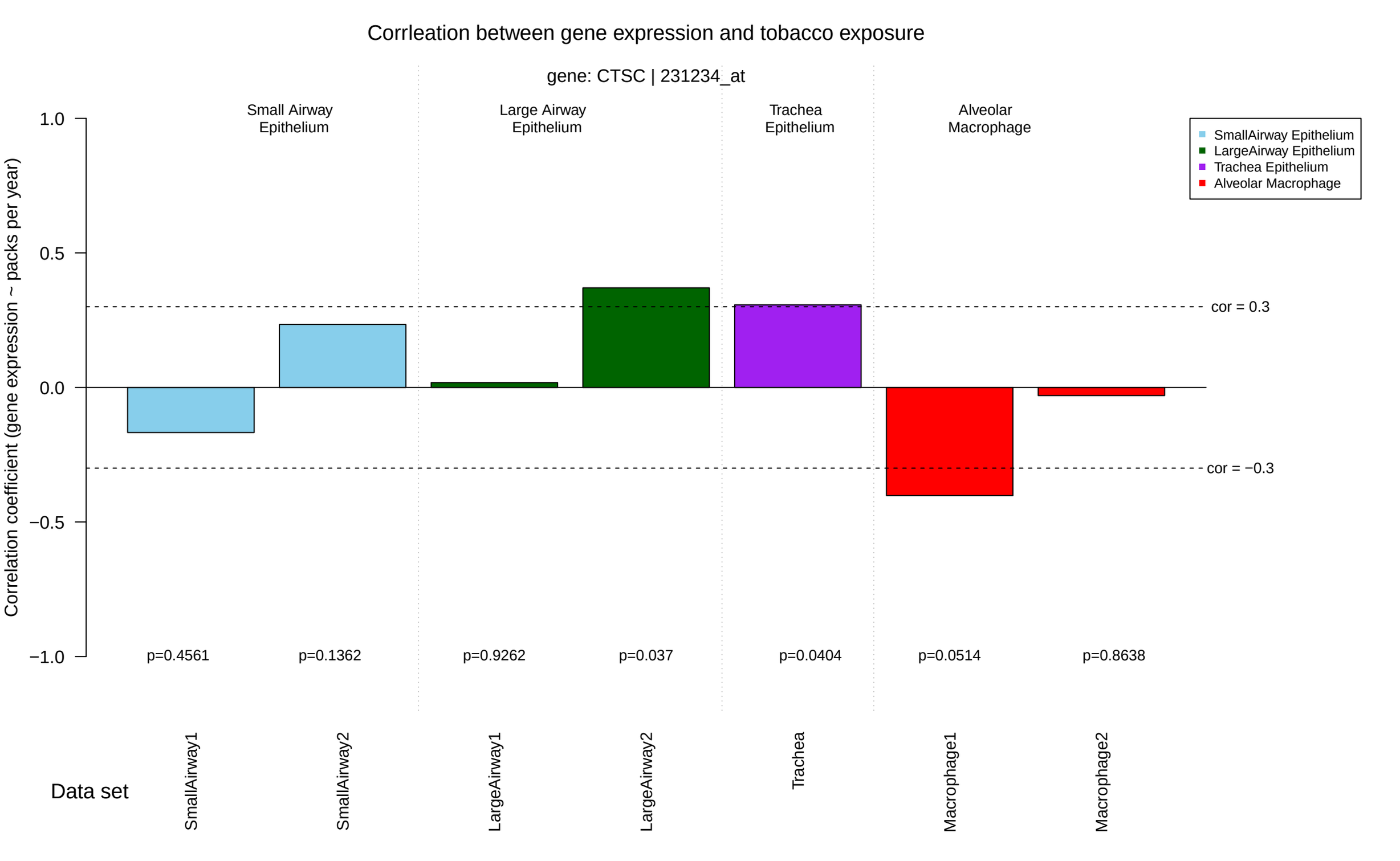

set: LargeAirway1 (p=0.9534)
CTSC | 231234_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6671)
CTSC | 231234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0391)
CTSC | 231234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0112)
CTSC | 231234_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6147)
CTSC | 231234_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CTSC | 231234_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4561
p=0.1362
p=0.9262
p=0.037
p=0.0404
p=0.0514
p=0.8638
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2